Unveiling the Financial Wellbeing Ecosystem: A Data-Driven Framework of Six Behavioral Profiles*

* Data Availability: The data used in this study are drawn from the 2021 National Financial Capability Study (NFCS), sponsored by the FINRA Investor Education Foundation. The dataset is publicly available at [https://finrafoundation.org/nfcs-data-and-downloads]

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Abstract

Traditional evaluations of financial wellbeing rely on unidimensional, linear indices that overlook the complex interaction between short-term preparedness and long-term security. This study proposes a novel intertemporal framework to capture the inherent heterogeneity of financial profiles through the integration of economic theory with machine learning techniques. Using data from the 2021 National Financial Capability Study (NFCS), a multi-stage pipeline is implemented, including theory-driven feature engineering, clustering and Random Forest classification to identify and validate six distinct segments, ranging from "The Established" to "The Distressed." Our findings reveal a consistent "Subjective Dominance" effect, where individual self-perception significantly outweighs objective metrics and cognitive factors in predicting wellbeing profiles. Furthermore, the analysis uncovers critical structural distinctions between vulnerable segments: while "The Short-Sighted" are primarily constrained by human capital deficits, "The Illiquid Planners" are destabilized by exogenous income shocks despite their planning capabilities. These results challenge the efficacy of generalized policy interventions and highlight the necessity of data-driven, psychologically informed strategies that account for the nuances of intertemporal trade-offs in personal finances.

Keywords – Financial wellbeing, financial literacy, financial fragility, financial stress

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ABSTRACT

Traditional evaluations of financial wellbeing rely on unidimensional, linear indices that overlook the complex interaction between short-term preparedness and long-term security. This study proposes a novel, intertemporal framework to capture the inherent heterogeneity of financial profiles through the integration of economic theory and machine learning techniques. Using data from the 2021 National Financial Capability Study (NFCS), a multi-stage pipeline is implemented, including theory-driven feature engineering, clustering, and Random Forest classification to identify and validate six distinct segments, ranging from “The Established” to “The Distressed”. The findings reveal a consistent “subjective dominance” effect, where individual self-perception significantly outweighs objective metrics and cognitive factors in predicting wellbeing profiles. Furthermore, the analysis uncovers critical structural distinctions between vulnerable segments: while “The Short-Sighted” are primarily constrained by human capital deficits, “The Illiquid Planners” are destabilised by exogenous income shocks despite their planning capabilities. These results challenge the efficacy of generalised policy interventions and highlight the need for data-driven, psychologically informed strategies that account for the nuances of intertemporal trade-offs in personal finances.

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1. Introduction

Financial wellbeing can be defined as being and feeling in control of personal expenses, and demonstrating short-term preparedness and long-term security. According to Sticha et al. (2023), it refers to managing money to meet daily demands, but also considering a retirement plan. This concept becomes noteworthy when both historical and recent events claim a strong relationship between socioeconomic status and ill-health, and even mortality. Evidence shows that general distress and mental illness are strongly driven by financial issues (Greene & Patil, 2023). Without proper management, these pathologies impact individuals' quality of life, leading to a cycle of negative impacts on household welfare (Jiménez-Solomon et al., 2024).

Financial wellbeing has heterogeneous perspectives towards its measurement and evaluation. A common definition is based on numerical scores. Although granular, these metrics represent an over-simplification of an individual's wellbeing, which indeed has multiple dimensions (Salignac et al., 2020). Furthermore, several efforts have also been made to evaluate its determinants. Objective metrics of financial profiles (e.g., income) were the first variables modelled, followed by the incorporation of subjective approximations (e.g., feelings towards such income). More recently, financial literacy has been included given its strong correlation with the previous two concepts, all of them modelled along with demographic features to control for contextual scenarios (Lusardi & Streeter, 2023).

The research landscape acknowledges a clear problem: financial wellbeing lacks an integral evaluation system, where its periodic nature and fundamental determinants are jointly recognised. Previous work aimed to develop compound scales with a multidimensional approach (Sticha et al., 2023). However, its subjective methodology complicates transparent interpretation. Authors such as Wagner and Walstad (2019) and Netemeyer et al. (2018) have formulated mathematical models to evaluate the effect of certain regressors on target variables. Nevertheless, not all critical determinants are considered, likely distorting the actual effect of the covariates.

This study aims to address the issue by designing and testing a holistic financial wellbeing framework. The proposed framework fulfils both theoretical and practical fragmentation in concept measurement and determinant evaluation. Accordingly, the focus is on establishing accurate profiles and analysing the reasons behind such differentiation. For this purpose, the study follows a non-experimental, quantitative methodology, supported by descriptive and predictive analyses. The central evaluation is conducted through machine learning techniques due to their strong capacity to model complex variable interactions within an interrelated system.

Recognising the integral nature of personal finances is essential for maintaining economic balance and mitigating risks of neglecting critical areas such as health and education. In addition, understanding the underlying drivers behind individuals' financial situations is paramount to forecasting effective decisions and policies. These findings hold universal relevance for a broad spectrum of stakeholders: from ordinary citizens striving for financial autonomy, to policymakers aiming to enhance public health, and non-governmental organisations (NGOs) dedicated to assisting underrepresented populations. Ultimately, this framework serves all stakeholders committed to strengthening the financial wellbeing of populations across decades and even generations.

2. Literature review

Human health comprises a broad concept where mental and physical welfare are the central objective (World Health Organization, 2021). In this sense, financial wellbeing has been extensively researched due to its impact on both health domains. Mental illness, chronic stress, and medical avoidance are some of the consequences of poor financial management (Spivak et al., 2019). For instance, in the United States, the effects of the COVID-19 pandemic revealed a surge in psychological pathologies among individuals experiencing income instability or job loss. Ringlein et al. (2024) found that, between 6 and 10 months following the onset March in 2020, the psychological distress levels of those affected by income shocks were 9% higher than their stable-income counterparts. This disparity increased to 11% between months 25 to 29, suggesting a chronic mental health deterioration in the long-term. This divergence implies that the psychological burden of financial stress outlasted government interventions, highlighting a latent public health crisis.

Based on the above, financial wellbeing has emerged as an essential indicator of social stability and progress (Brüggen et al., 2017). While its preservation is a central objective globally, heterogeneous conceptualisations and fragmented operationalisations continue to obscure a transparent understanding (Mahendru et al., 2022). In this sense, a robust measurement and analysis of financial wellbeing is critical to mitigate systemic public health risks and foster long-term resilience (Salignac et al., 2020).

Regarding conceptualisation, the literature presents a range of perspectives drawing on economic foundations, technical definitions, and heuristic analogies, as discussed by Brüggen et al. (2017), Sorgente, Totenhagen, and Lanz (2022), and Sajid et al. (2024). These multiple approaches overcomplicate a general understanding. To overcome this issue, Sticha, Lusardi, and Sconti (2023) proposed a comprehensive definition, stating that "financial wellbeing is not only about long-term financial security (retirement planning) but also about short-term financial preparedness". This contribution reinforces a previous approximation of Salignac et al. (2020), who concluded that a person is financially healthy when "they are in control of their finances and feel financially secure, now and in the future".

The previous foundation is critical for providing a complete measurement. Thus, the intertemporal duality of financial wellbeing is employed in this study, defining it as the effective management of day-to-day expenses and proper budget planning for the future. As noted above, current measurements are solely based on one-dimensional scores or subjective compound metrics. Crucially, the majority of existing indices do not assess the intertemporal integrability of financial wellbeing. These limitations increase the risk of misguided conclusions and ineffective interventions. Given the above, the first research question is formulated as follows:

RQ1: How can financial wellbeing be measured within an intertemporal framework?

The previous methodological heterogeneity is not limited to its measurement; it also extends to the analysis of the drivers behind financial wellbeing outcomes. Research on household financial wellbeing has normally been conducted considering income-based drivers, plus demographic controls (Cardona-Montoya et al., 2022). For instance, studies such as Theodossiou (1998) were focused primarily on the impact of low wages on mental health. Twenty years later, Dolan, Peasgood, and White (2008) presented an interesting contribution stating that happiness was more strongly influenced by relative income comparisons than by the actual income received by individuals. In other words, self-assessments towards one's own situation represent an indispensable factor on personal welfare. This finding explains the origin of the subjective-objective dichotomy and adds a second driver of financial wellbeing.

Notably, a common feature among existing works is the acknowledgement of structural differences reflected in demographic variables. In this sense, Wagner and Walstad (2019) suggest that both momentary and long-lasting circumstances must be considered, underscoring the significance of age, gender, income range, and marital status, among other variables.

More recently, Lusardi and Streeter (2023) concluded that financial literacy plays a fundamental role in shaping money-management practices (subjective perceptions) and financial decision-making (objective behaviour), finding a strong impact on financial wellbeing. This third variable complemented the set of determinants.

Although several studies have tested the effects of these three determinants, they were mostly analysed in isolation. In this sense, financial wellbeing research still lacks an integral modelling where all the aforementioned drivers are jointly evaluated (subjective perceptions along with objective behaviours and financial literacy). By providing a robust model, a better understanding of financial wellbeing outputs, their causes and consequences is achieved. This contributes to conscious decisions and tailored actions. Given the above, the second research question is formulated as follows:

RQ2: What is the role of the entire set of financial wellbeing determinants?

To address these research questions, this study adopts a robust quantitative approach. Specifically, to answer RQ1, an index construction is constructed for each intertemporal dimension, followed by its consolidation into a comprehensive framework that presents a novel taxonomy of financial wellbeing profiles. Subsequently, to address RQ2 and explore the interplay between determinants, a machine learning algorithm is trained, and a granular analysis is conducted on the role of each determinant.

3. Method

The strategy employed to address the research questions follows a sequential, multi-stage architecture. This is structured into three phases to ensure robustness and reproducibility. First, the research design and data sourcing are established to guarantee sample accuracy and representativeness (Section 3.1). Then, a rigorous feature engineering is conducted to operationalise the intertemporal measurements and determinants of financial wellbeing (Section 3.2). Finally, the implementation and testing of the framework are detailed, describing the machine learning technique used to uncover the novel taxonomy (Section 3.3).

3.1. Research design, sample and data

The methodology comprises a quantitative and non-experimental analyses, with a strong focus on variable interactions within the created framework. This research is based on the 2021 National Financial Capability Study (NFCS), a strategic dataset to analyse the COVID-19 pandemic effects on population financial wellbeing (FINRA Investor Education Foundation, 2021). It comprises information of 27,118 adults from the United States, representing a population of over 250 million. Of those, all observations which belong to retired individuals or lacking responses to key questions to estimate the model variables were excluded. In this sense, the final sample involves 11,857 observations. The strength of this dataset relies on its vast amount of information for a complete analysis. Financial behaviours, perceptions, and education are covered, along with sociodemographic and contextual features.

3.2. Variable construction and feature engineering

Based on the literature review, the dependent variable was built within an intertemporal framework. Following the method adapted from Wagner and Walstad (2019), eight items from the NFCS were selected to construct this measure. Specifically, four items are designated to assess short-term (ST) financial control, while the remaining four evaluate long-term (LT) financial security.

To construct these two indices, a feature engineering process is employed, involving the binarisation and additive aggregation of the selected items. While dimensionality reduction techniques, such as Principal Component Analysis (PCA) or Multiple Correspondence Analysis (MCA), are common in large datasets, feature engineering was preferred to preserve the theoretical interpretability and semantic integrity of the financial constructs. Unlike PCA, which generates latent components that maximise statistical variance but often lack direct interpretation (Zytek et al., 2022; Karaahmetoğlu et al., 2025), this approach ensures that the resulting indices directly reflect the intertemporal duality of financial wellbeing, as defined in the literature. The detailed interpretation of the drivers, which purely statistical components might obscure, is key for actionable insights (Huston, 2010). In this sense, binarisation and aggregation enable the synthesis of variables from raw data while maintaining consistency with the objectives and preserving the dependencies within the model (Kuhn & Johnson, 2013).

The short-term index is constructed using four questions shown in Table 1. These items are designed to capture the immediate cash-flow management and daily financial behaviours that define an individual's financial stability. Following the framework proposed by Wagner and Walstad (2019), four key dimensions were identified: expense management, banking behaviour, credit discipline, and debt perception. Each item was binarised (1 = favourable, 0 = unfavourable) using a "top-box" criterion to isolate optimal financial behaviours (such as paying credit cards in full or having no difficulty covering bills) from any degree of financial distress. This operationalisation allows the index to reflect clear distinctions between financial stability and vulnerability in the short-term.

Table 1

Operationalisation and binary encoding of short-term financial wellbeing components

Construct	NFCS item	Binarisation logic (1 = favourable)
Expense management	J4: In a typical month, how difficult is it for you to cover your expenses and pay all your bills?	Not at all difficult
Banking behaviour	B4: Do you [or your spouse/partner] overdraw your checking account occasionally?	No (Never occasionally overdraws)
Credit discipline	F2_1: In the past 12 months, which of the following describes your experience with credit cards? - I always paid my credit cards in full.	Yes (Always paid in full)
Debt perception	G23: How strongly do you agree or disagree with the following statement? - I have too much debt right now.	Strongly disagree/Disagree

Note: Data from the 2021 National Financial Capability Study (NFCS).

The long-term index is measured by four questions shown in Table 2. These items focus on asset accumulation and future resilience, adapting the methodology of Wagner and Walstad

(2019) as well. They cover emergency resilience and formal liquidity, essential metrics to avoid financial stress during economic downturns (Haveman & Wolff, 2005). Furthermore, investment activities and retirement planning items are considered, which are critical for life-cycle financial wellbeing (Lusardi et al., 2017). Consistent with the short-term index, all items are binarised using a threshold of proactive financial security. Favourable outcomes reflect anticipatory financial actions rather than mere asset ownership. For example, retirement planning is defined by the active calculation of future needs rather than merely the existence of a retirement account.

Table 2

Operationalisation and binary encoding of long-term financial wellbeing components

Construct	NFCS item	Binarisation logic (1 = favourable)
Emergency resilience	J5: Have you set aside emergency or rainy-day funds that would cover expenses for 3 months?	Yes (Has 3-month buffer)
Liquidity/basic saving	B2: Do you have a savings account, money market account, or CDs?	Yes (Maintains formal savings)
Asset accumulation	B14: Not including retirement, do you have any investments in stocks, bonds, or mutual funds?	Yes (Active investor)
Retirement planning	J8: Have you ever tried to figure out how much you need to save for retirement?	Yes (Has calculated retirement needs)

Note. Data from the 2021 National Financial Capability Study (NFCS).

Subsequently, the independent variables are defined, representing the entire set of critical determinants of financial wellbeing. This approach comprises objective behaviours, subjective perceptions, financial literacy, and demographic characteristics. Twenty-six items were selected from the NFCS to construct these measures. Specifically, five items were designed to assess the objective behaviour, five to evaluate the subjective perceptions, seven to measure financial literacy, and nine to reference the individuals' demographic characteristics.

The objective behaviour index is measured by the questions shown in Table 3. These items serve as a proxy for personal pillars for financial stability and security. The inclusion of the income threshold at the $75,000 mark aligns with established literature suggesting that emotional stability tends to plateau beyond this income level (Kahneman & Deaton, 2010). Budgetary discipline is incorporated as a measure of active saving behaviour, reflecting the individual's capacity for capital retention regardless of total earnings. Liquidity shock capacity represents the NFCS standard for assessing financial fragility. By isolating only those who are "totally confident," this index applies a stringent filter for immediate financial resilience (Lusardi et al., 2011). Finally, the index accounts for institutional coverage and credit risk avoidance, capturing the external support systems and the avoidance of high-cost, unsustainable debt.

Table 3

Operationalisation and binary encoding of objective behaviour components

Construct	NFCS item	Binarisation logic (1 = favourable)
Income threshold	A8_2021: What is your household's approximate annual income?	1 = high income (> $75,000)
Budgetary discipline	J3: Was your household's spending less than, more than, or equal to your income?	1 = budget surplus (spending < income)
Liquidity shock capacity	J20: How confident are you that you could come up with $2,000 for an unexpected need?	1 = high liquidity confidence (totally confident)
Institutional coverage	C1_2012: Do you have any retirement plans through an employer (e.g., pension)?	1 = covered (has employer-sponsored plan)
Credit risk avoidance	G25_1: In the past 5 years, how many times have you taken out an auto title loan?	1 = safe credit behaviour (never)

Note: Data from the 2021 National Financial Capability Study (NFCS).

The subjective perception index is measured by the questions shown in Table 4. The abbreviated version of the Consumer Financial Protection Bureau (CFPB) scale was employed for this purpose. This validated index captures the psychological dimension of financial wellbeing, moving beyond mere solvency to measure individuals' internal sense of security and freedom. Unlike the previous indices, these items were not binarised; instead, a two-step transformation process was applied. First, raw responses were coded on a 0–4 scale depending on their wellbeing phrasing. Second, the cumulative scores were mapped onto the official CFPB Conversion Table to produce a final standardised score ranging from 0 to 100. Following the CFPB technical protocol, negatively phrased items were reverse-coded to ensure that higher values consistently reflect greater levels of perceived wellbeing.

Table 4

Operationalisation and binary encoding of subjective perception components

Construct	NFCS item	Scoring methodology
Long-term goal attainment	*How well do these statements describe you?* Because of my money situation, I feel like I will never have the things I want in life.	Negatively phrased 0 = completely 4 = not at all
Financial sufficiency perceptions	*How well do these statements describe you?* I am just getting by financially.	Negatively phrased 0 = completely 4 = not at all

Future security anxiety	*How well do these statements describe you?* I am concerned that the money I have or will save won't last.	Negatively phrased 0 = completely 4 = not at all
Monthly budgetary margin	*How often do these statements apply to you?* I have money left over at the end of the month.	Positively phrased 4 = always 0 = never
Perceived financial agency	*How often do these statements apply to you?* My finances control my life.	Negatively phrased 0 = always 4 = never

Note: Data from the 2021 National Financial Capability Study (NFCS).

The financial literacy index is measured by the questions shown in Table 5. This index measures the cognitive foundation of financial decision-making. Following the work of Lusardi and Mitchell (2011), the index incorporates the "Big Three" concepts (interest rates, inflation, and risk diversification), alongside more advanced topics such as bond pricing, mortgage structures, compound interest, and probabilistic reasoning. These items serve to differentiate between basic numeracy and functional financial knowledge. Each item was binarised (1 = correct, 0 = incorrect/do not know), creating a cumulative scale that ranges from 0 to 7.

Table 5

Operationalisation and binary encoding of financial literacy components

Construct	NFCS item	Binarisation logic (1 = correct answer)
Numeracy/interest	Suppose you had $100 in an account with an interest rate of 2% per year. How much would you have after five years if you left the money to grow?	1 = more than $102
Inflation awareness	Imagine that the interest rate on your account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money?	1 = less than today
Risk diversification	Buying a single company's stock usually provides a safer return than a stock mutual fund.	1= false
Bond valuation	If interest rates rise, what will typically happen to bond prices?	1 = they will fall
Mortgage literacy	A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less.	1= true
Compound interest	Suppose you owe $1,000 on a loan and the interest rate is 20% per year compounded annually. If you didn't	1 = 2 to 5 years

	pay anything off, at this interest rate, how many years would it take for the amount you owe to double?	
Probabilistic reasoning	Which of the following indicates the highest probability of getting a particular disease?	1 = one-in-twenty

Note: Data from the 2021 National Financial Capability Study (NFCS).

Lastly, nine demographic and socioeconomic variables are integrated as control features to account for individual heterogeneity in financial outcomes. These include age group, gender, educational attainment, household income, number of dependent children, income shocks, secondary employment, labour status, and marital status. All of them were extensively employed in previous work, such as the studies of Lusardi and Streeter (2023), Sticha, Lusardi, and Sconti (2023), and Wagner and Walstad (2019). The ordinal structure of age, education, income, and number of dependants was preserved to capture the inherent gradients in these dimensions. Conversely, the nominal variables were all transformed using one-hot encoding. This process generates distinct binary indicators for each category, preventing the model from assuming a non-existent hierarchy among nominal labels. A detailed description of each indicator and its corresponding measurement scale is provided in the Online Appendix (Fig. B.1).

3.3. Implementation of the financial wellbeing framework

Based on the previously constructed indices, the dependent variable was derived from a cross-tabulation of the short-term (ST) and long-term (LT) scales. As illustrated in Fig. 1, this bivariate approach maps the intertemporal framework of financial wellbeing, resulting in a 5x5 matrix with twenty-five distinct cells. This spatial representation allows for a granular identification of financial profiles. The bottom-left quadrant identifies individuals experiencing concurrent fragility in both ST and LT dimensions, whereas the top-right quadrant characterises those exhibiting strong intertemporal security. Consequently, this matrix serves as the structural basis for operationalising the financial wellbeing profiles, thereby defining the model's dependent variable.

With both the dependent and independent variables constructed, a supervised classification model was implemented to simultaneously validate the proposed framework and analyse the financial wellbeing of U.S. adults. Two models are estimated for this purpose. Equation (1) represents the model without control variables, and Equation (2) adds the demographic controls. Both models are presented below:

$$Y_i = f(\mathrm{X}_{1,\mathrm{i}} + \mathrm{X}_{2,\mathrm{i}} + \mathrm{X}_{3,\mathrm{i}}) + \epsilon_{\mathrm{i}} \tag{1}$$

$$Y_i = f(\mathrm{X}_{1,\mathrm{i}} + \mathrm{X}_{2,\mathrm{i}} + \mathrm{X}_{3,\mathrm{i}} + \boldsymbol{Z_i}) + \epsilon_{\mathrm{i}} \tag{2}$$

where:

Y_i is the dependent label value for individual "*i*"
$\mathrm{X}_{1,\mathrm{i}}$ is the independent index value of subjective perception for individual "*i*"
$\mathrm{X}_{2,\mathrm{i}}$ is the independent index value of objective behaviour for individual "*i*"

$X_{3,i}$ is the independent index value of financial literacy for individual "*i*"
Z_i is the set of demographic controls for individual "*i*"
ϵ_i is the random error term for individual "*i*"

The algorithm selected is Random Forest (RF) due to its superior capacity to handle heterogeneous data types and capture the nonlinear relationships inherent in socioeconomic data (Wright & König, 2019). As an ensemble method that aggregates multiple decision trees, RF is robust against outliers and multicollinearity (Kuhn & Johnson, 2013), making it suitable for the complex behavioural interactions within financial datasets. By adopting this approach, the study provides an empirical validation of the bi-dimensional framework presented in Fig. 1. The dataset is then partitioned into training (80%) and testing (20%) subsets to ensure out-of-sample evaluation. To address class imbalance, the Synthetic Minority Over-Sampling Technique (SMOTE) is applied exclusively to the training sample.

The Random Forest implementation in this study follows a principled approach for hyperparameter tuning (Breiman, 2001; James et al., 2023). The number of trees is set to 100, sufficient for the out-of-bag error to converge. The number of features at each node split is set to the square root of the number of predictors, a common heuristic to reduce correlation between trees. These choices target the main sources of ensemble variance and bias. The remaining hyperparameters were kept at their default values to avoid overfitting and ensure a reproducible baseline framework.

Model performance was evaluated using a robust set of out-of-bag measure error estimates. While global accuracy and the confusion matrix provided a baseline for overall model validity, the analysis prioritised precision and recall (sensitivity) to ensure a proper assessment of the classifier's performance across heterogeneous segments. Given the inherent class imbalance within the financial wellbeing clusters, these last two metrics were prioritised for minority classes. This multi-metric approach ensures that the framework's predictive power is consistently validated, even for underrepresented populations within the NFCS dataset. Furthermore, the inclusion of feature importance metrics enables an analysis of the primary drivers of financial wellbeing, ensuring that the model fulfils both the classificatory and diagnostic objectives of this research.

4. Results

The development of the intertemporal framework reveals a complex landscape of financial wellbeing that linear models often obscure. This section presents the findings in three stages. First, a novel, six-clusters taxonomy is established to measure the U.S. population across the short-term and long-term spectrum. Second, the most important determinants in shaping financial wellbeing are identified through the Global Random Forest classification. Finally, an original examination of "boundary dynamics" is presented, identifying the specific drivers that differentiate adjacent clusters, thus offering granular insights into the mechanisms of financial mobility.

4.1. New taxonomy of financial wellbeing

Fig. 1 shows that the intertemporal framework is not distributed uniformly. While a significant density of observations (n) clusters in the high-performance quadrant (indicating intertemporal synergy), distinct pockets of vulnerability emerge in the lower bounds and off-diagonal axes. Following the segmentation logic applied by Azevedo et al. (2024) in social science research, this study prioritises a clear distinctiveness over equally sized groupings. The objective is not to create statistical quartiles, but to isolate relevant profiles that exhibit contrasting financial realities. Consequently, the 25-cell matrix was synthesised into six distinct clusters, forming a new taxonomy of financial wellbeing.

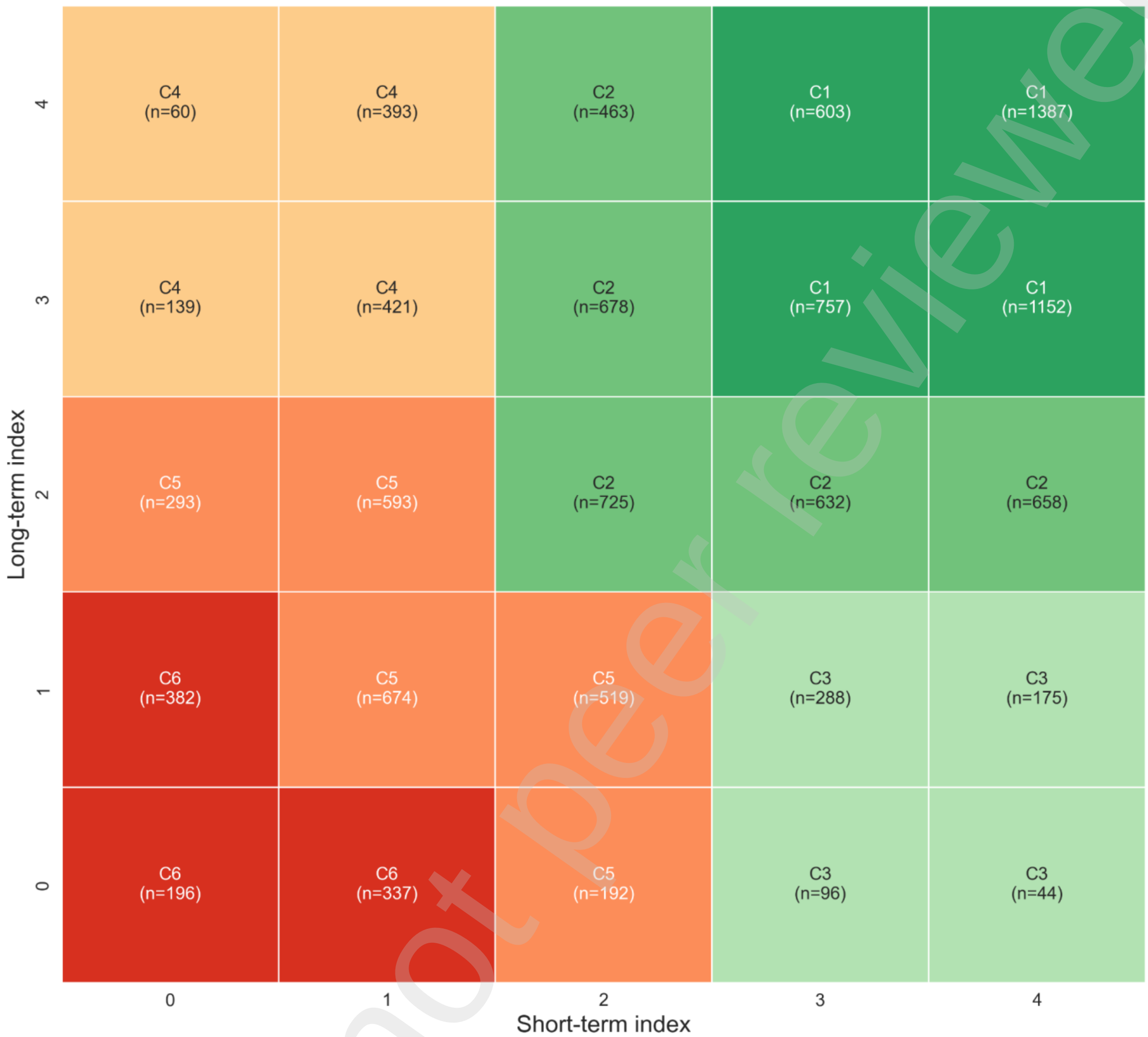

Fig. 1. The intertemporal financial wellbeing framework: identification of six distinct profiles. Note: The framework illustrates the six distinct profiles identified through the intersection of the short-term and long-term indices.

Drawing on the spatial configuration presented above, the proposed framework delineates distinct profiles with unique intertemporal combinations, ranging from robust stability to systemic vulnerability. The specific cluster characteristics are detailed below.

Cluster 1: The Established (C1). This group reaches the highest level of financial health. It is characterised by strong liquidity management in the present and robust capital planning for the future.

Cluster 2: The Resilient (C2). This segment demonstrates moderate control over current finances and adequate future security, but it has sub-optimal financial habits.

Cluster 3: The Short-Sighted (C3). This cluster maintains adequate liquidity and consumption satisfaction but shows deficient long-term preparedness.

Cluster 4: The Illiquid Planners (C4). In contrast to C3, this group has managed to construct a substantial future budget, but struggles to fund its day-to-day demands.

Cluster 5: The Precarious (C5). This segment presents a latent risk, marked by instability and lack of control of their finances. They are theoretically solvent, but vulnerable in practice to any kind of income shock.

Cluster 6: The Distressed (C6). This segment occupies the lower tier of financial health, revealing immediate liquidity crises and minimal safety for the future.

The descriptive analysis of the three core determinants across the six identified clusters is presented in Fig. 2. A progressive gradient is evident. Cluster 1 (C1) exhibits the highest mean scores across all dimensions, justifying its label as The Established, while the indicators steadily decrease until reaching the lowest scores within The Distressed (C6). Intermediate clusters reveal distinct differences in determinants, yet substantial heterogeneity in outcomes. A more granular analysis of this critical differentiation is presented in Section 4.3.

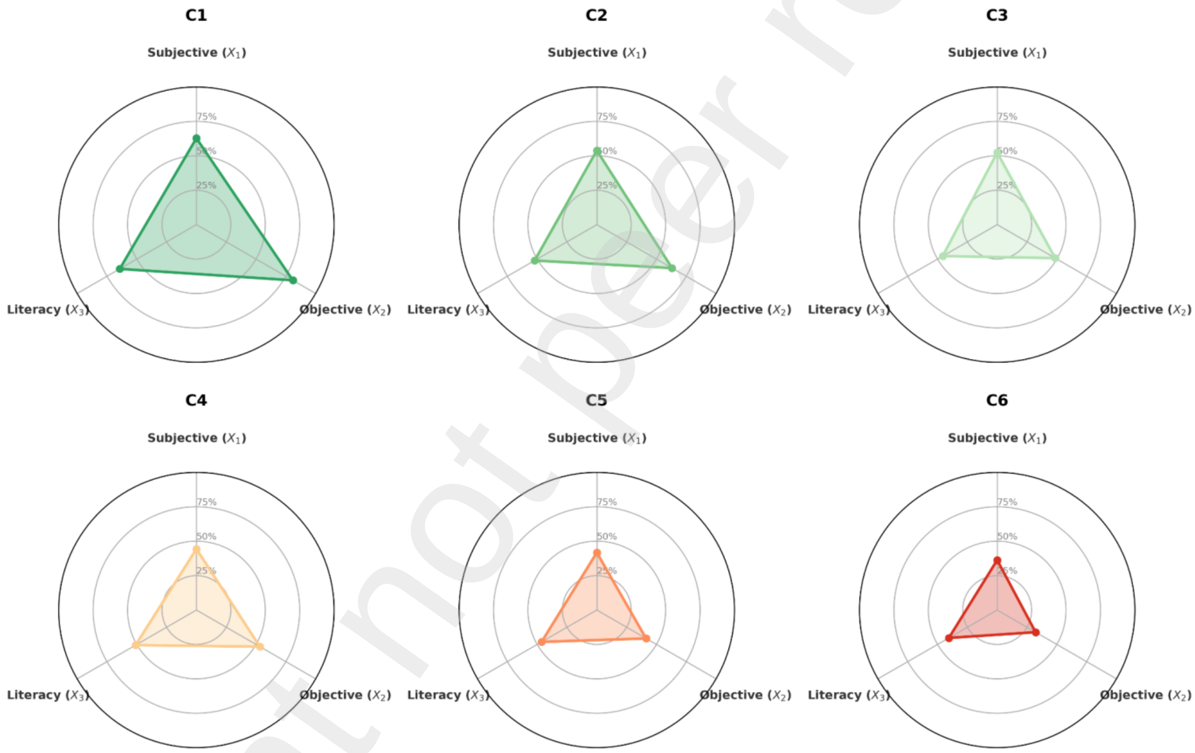

Fig. 2. Multidimensional profiling of financial wellbeing clusters: a comparative analysis of key determinants. Note: The chart displays the normalised average scores for each cluster, rescaled to 0-100% relative to the theoretical range of each index. Values closer to the periphery indicate a higher prevalence of the attribute.

Beyond the heterogeneity evidenced in Fig. 1, clear socioeconomic and structural differences emerge among the six clusters, and their acknowledgement supports a granular understanding of such profiles. Detailed distributions of these demographic characteristics are presented in Online Appendix B. Education and income among The Established (C1) play a fundamental role within this segment. A significant portion of individuals with postgraduate degrees (47.1%) and individuals with annual earnings above \$300,000 (72.5%) are located in C1. In contrast, within The Distressed (C6), 79.7% have not attained an undergraduate degree and 66.1% earn less than \$50,000 annually (Figs. B.2-B.3). A gender gap is also evident.

Females are disproportionately represented in the less healthy segment, comprising 69.4% of The Distressed, whereas males dominate the most-healthy cluster reaching 57.8% of The Established (Fig. B.4).

Regarding the lifecycle hypothesis of Wagner and Walstad (2018), financial wellbeing generally improves with age, suggesting that financial maturity and prudence develop over time (Fig. B.5). An exogenous incident, based on the COVID-19 pandemic effects, was critical for group demarcation. Only 12% of The Established reported an income disruption, compared with an average of 48.6% among the vulnerable clusters (The Illiquid Planners, The Precarious, and The Distressed) (Fig. B.6).

Finally, family composition and marital status indicate no global patterns; however, they are relevant for clusters with intertemporal biases, such as The Short-Sighted (C3) and The Illiquid Planners (C4). There is a significantly higher proportion of dependent children within C4, compared with C3, reaching a difference of 31.6%. This suggests that family demands constrain liquidity for daily expenses. Moreover, The Short-Sighted, largely composed by non-partnered individuals (59.7% belong to single, separated, divorced, or widowed statuses), may direct income flows toward required immediate consumption, preventing long-term savings (Figs. B.7-B.8).

4.2. Global determinants of financial wellbeing

After the initial diagnostic analysis, the stepwise Random Forest application provides both confirmatory and novel findings. Two models are estimated: Model 1 is the baseline, with only the three core determinants (objective, subjective, and literacy scales), while Model 2 adds the sociodemographic controls. The results, summarised in Table 6, show an expected insight: the inclusion of contextual and personal characteristics is critical to capture the variance in financial wellbeing outcomes. Model 2 improves the accuracy from 0.37 to 0.46, with consistent gains in precision and recall metrics. This enhanced performance validates previous studies, such as Wagner and Walstad (2019), Lusardi and Streeter (2023), and Sticha, Lusardi, and Sconti (2023), which established that an individual's contextual environment deeply influences their financial wellbeing.

Table 6

Comparative performance of Random Forest models: baseline determinants vs. full demographic model

Metric	Model 1	Model 2
Accuracy	0.3689	0.4557
Precision		
C1	0.70	0.66
C2	0.38	0.38
C3	0.08	0.14
C4	0.15	0.27
C5	0.33	0.40
C6	0.28	0.31

Metric	Model 1	Model 2
Recall		
C1	0.64	0.70
C2	0.18	0.31
C3	0.23	0.12
C4	0.32	0.33
C5	0.20	0.44
C6	0.46	0.34

Note: Accuracy indicates overall model performance, whereas precision and recall provide a granular assessment of each cluster, controlling for class-size disparities.

Beyond the classification metrics, the feature importance analysis provides surprising findings into the determinants structure. Table 7 presents a ranking analysis for both models. Even though there is a slight variation in variable significance between the two models, the Subjective Index emerges as the paramount classifier, exhibiting superior predictive importance compared with the Objective Index and the Financial Literacy Index. This hierarchy evidences that individuals' internal sense of preparedness and security (their mindset) is more important than their actual financial behaviours and habits (their actions). Regarding the demographic features, results for age and education level confirm the previous descriptive analyses, indicating that financial health involves life-cycle dynamism, with education acting as a strong differentiator.

Table 7

Random Forest feature importance (FI): hierarchy of financial wellbeing determinants

FI Ranking	Model 1	Model 2
1	Subjective Index	Subjective Index
2	Objective Index	Literacy Index
3	Literacy Index	Objective Index
4		Education Level
5		Age Group

Note: Higher ranking indicates a greater predictive contribution of the variable in distinguishing financial wellbeing profiles.

The inclusion of sociodemographic controls allows the model to classify with higher accuracy. Nevertheless, the confusion matrix shown in Table 8 reveals persistent misclassification between adjacent clusters. Particularly, individuals among the intermediate clusters, The Short-Sighted (C3), The Illiquid Planners (C4), and The Precarious (C5), are likely to overlap because of their similar characteristics. Likewise, The Established (C1) are frequently misclassified as The Resilient (C2), and vice versa. This phenomenon was expected, as previous work suggests that financial wellbeing modelling is indeed complex, where similar determinants may result in different outcomes (Dolan et al., 2008; Lusardi & Streeter, 2023). Ultimately, the heterogeneity among individual profiles within each cluster underscores that there is not a unique or infallible determinant, neither at the personal nor at the contextual level.

Table 8

Confusion matrix: analysis of inter-cluster misclassification patterns

Model 1							Model 2						
	C1	*C2*	*C3*	*C4*	*C5*	*C6*		*C1*	*C2*	*C3*	*C4*	*C5*	*C6*
C1	498	102	87	55	31	7	*C1*	546	151	13	29	33	8
C2	168	111	125	128	55	44	*C2*	193	194	44	71	105	24
C3	18	14	28	35	14	12	*C3*	28	26	14	6	37	10
C4	14	25	27	64	46	27	*C4*	26	48	2	66	46	15
C5	15	33	80	110	89	127	*C5*	29	69	22	56	199	79
C6	3	8	25	30	32	85	*C6*	6	18	5	18	74	62

Note: Rows represent the actual (true) cluster membership, while columns represent the predicted classification by both models.

4.3. The dynamics of mobility: boundary analysis

Previous analyses suggest that global determinants delineate the broad framework for financial wellbeing. However, they obscure the specific drivers that prevent social mobility across adjacent clusters. To identify these “levers of change”, a new strategy is adopted by isolating the decision boundaries between key segments. This is implemented through three binary Random Forest classifications, the results of which are presented in Table 9. The granular analysis reveals that financial wellbeing determinants shift according to the location of the individual within the intertemporal framework.

Table 9

Pairwise discriminant analysis: evaluation metrics across adjacent clusters

Metric	Pair C3 - C4	Pair C5 - C6	Pair C1 - C2
Accuracy	0.7932	0.6426	0.7030
Feature importance			
1	Subjective Index	Subjective Index	Subjective Index
2	Income Shock	Literacy Index	Literacy Index
3	Literacy Index	Education Level	Objective Index
4	Education Level	Objective Index	Age Group
5	Dependent Children	Age Group	Education Level

Note: Results are derived from distinct Random Forest applications on each specified pair of clusters. Higher ranking of feature importance indicates a greater predictive contribution of the variable in distinguishing financial wellbeing profiles.

The "mismatch analysis" behind The Short-Sighted (C3) and The Illiquid Planners (C4)

The intertemporal imbalance of these clusters indicates a critical policy insight. Both segments present bias, however, the discrimination is not entirely behavioural, but event-driven. The binary classification (accuracy: 0.79) reveals that they represent structurally different populations influenced by opposing forces.

The Random Forest identifies Subjective Index and Income Shock as the top-tier differentiators, as shown in the first column of Table 9. Notably, 58% of The Illiquid Planners experienced a recent income disruption, compared with 16.6% of The Short-Sighted. This suggests that C4 weakness does not lie in poor financial habits, but rather in liquidity hardship due to exogenous events, including job loss and business failure. This erosion of C4 short-term stability, despite their long-term assets, is also impacting on their self-assessment, leading to an unusual decline in its Subjective Index. Conversely, the friction in The Short-Sighted is structural and, mainly, educational. Only 28.9% of C3 hold a bachelor's degree or higher, leading to a human capital gap compared with their same-age peers. Their lack of future preparedness is also exacerbated by their low Financial Literacy Index, meaning a significant systematic risk. This distinction implies that C4 requires safety nets such as insurance, emergency liquidity, or temporary government funding, while C3 requires structural interventions in financial literacy and general education.

The "survival gradient analysis" behind The Precarious (C5) and The Distressed (C6)

Analysing the vertical transitions at the framework extremes reveals a different dynamic. At the bottom of the framework, comprising C5 and C6, the distinction is driven by a "poverty trap" mechanism. Fig. 2 and the second column of Table 9 show that The Distressed are separated from The Precarious by a set of aggravated negative indicators. They exhibit consistently lower financial literacy, lower general education, more negative perceptions, and, crucially, greater exposure to income shocks (49.2% vs. 38.7%). In this low-level zone, mobility is largely shaped by an accumulation of disadvantages, likely exacerbated by the consequences of COVID-19.

The "optimisation gradient analysis" behind The Resilient (C2) and The Established (C1)

At the top of the framework, composed of C2 and C1, the driver of differentiation shifts from "survival" to "optimisation". The boundary between The Established and The Resilient is not driven by shocks or income alone, but by marginal gains resulting from cognitive and maturity alignment, as shown in Fig. 2 and the third column of Table 9. C1 individuals exhibit a "synergistic advantage" with higher literacy (+2.03 points), markedly better subjective perception (+8.88 points), and older age (maturity effect). Specific details regarding this differentiation are presented in the Online Appendix (Fig. B.9). This suggests that transitioning from "middle class" to "wealthy established" is an evolutionary process driven by the accumulation of assets and confidence over the lifecycle, with financial education playing a significant mediating role.

5. Discussion

The empirical identification of six distinct profiles challenges the traditional view of financial wellbeing as a linear and unidimensional construct. By revealing the structural heterogeneity among U.S. households, this study demonstrates that financial wellbeing is driven by the dynamic interplay between present stability and future security. The following sections discuss the implications of these findings: first, the theoretical necessity of transcending to robust, comprehensive frameworks (Section 5.1); and second, the practical shift from generic interventions to tailored and data-driven policies (Section 5.2).

5.1. Theoretical implications: beyond the linear paradigm

This study expands the literature on financial wellbeing by shifting the analytical scope from linear and independent models to a multidimensional and intertemporal framework. Previous research has largely employed continuous gradients (Lusardi & Streeter, 2023; Sticha et al., 2023); however, this study confirms that financial wellbeing is better understood as a heterogeneous system of behavioural-structural clusters.

Specifically, this analysis provides strong support for Netemeyer et al. (2018) regarding the importance of subjective perceptions. A critical insight from this study is the "subjective dominance effect", whereby an individual's self-assessment plays a pivotal role in their position within the framework, even more than their objective behaviours or literacy levels. This suggests that psychological strain is not merely an outcome of financial wellbeing, but a structural driver that can shift individuals into lower wellbeing clusters or propel them upward. This creates a feedback loop in which negative perceptions, aggravated by income shocks, can paralyse decision-making, even among literate individuals, as observed in The Illiquid Planners (C4). The previous insight complements and expands the cyclical phenomenon of financial wellbeing and mental health acknowledged by Jiménez-Salomon et al. (2024).

Furthermore, this research enriches the foundational work of Lusardi and Streeter (2023) and Wagner and Walstad (2019). Although the well-known "Big Three" questions used to measure financial literacy constitute a validated metric, this study demonstrates that expanding the scale to capture more complex concepts is critical. A novel finding of this research is that, while basic literacy is necessary for shifts between intermediate clusters (e.g., from C5 to C4), mastery of advanced financial knowledge acts as the gatekeeper to the highest wellbeing tiers (C1 shows a significant, unusual leap in its Literacy Index compared with C2 and the other clusters). Similarly, the present-future dichotomy in financial wellbeing measurement cannot be analysed in isolation as it obscures the intertemporal foundation for individuals profiling. This research highlights that a multidimensional analysis provides valuable insights for uncovering latent risks and optimisation opportunities, which are key to effective policy design.

5.2. Policy and managerial implications: from generic to precise decision-making

The heterogeneity revealed by the six-cluster framework classification argues against a unique, conventional policy intervention. Effective decisions require a data-driven triage tailored to the specific needs of each group. Based on the insights of this study, a three-tiered policy intervention is proposed.

Tier 1: The structural-based intervention for the vulnerable segments: C5 and C6

The Distressed (C6) and The Precarious (C5) require an income stabilisation mechanism and social protection strategies, rather than purely literacy programmes, which are likely to be insufficient and misdirected. Data indicate that their financial fragility is driven by a systematic lack of resources, rather than an absence of a money-management guide. For individuals in this segment, food security and humanitarian transfers are essential to prevent a compounding effect of multidimensional deprivation (Lee, 2023). With the aim of avoiding a positional drop from C6 to C5, educational interventions are also needed, focusing on survival financial management, where fraud prevention and excessive debt avoidance should be prioritised given their vulnerable social situation.

Tier 2: The equilibrium-based intervention for the middle segments: C3 and C4

Distinguishing The Short-Sighted from The Illiquid Planners is a significant contribution that this research offers to policymakers for effective intervention.

- For The Short-Sighted (C3), their deficit in both general and financial education demands priority literacy interventions through mandatory curricular modifications with a strong focus on long-term financial awareness. Conversely, the adults who are no longer part of the traditional education system, a commitment contract strategy is an effective tool to "force" individuals to enrol in saving plans and counteract their present bias (Halpern et al., 2012).
- For The Illiquid Planners (C4), their deficit in contingency planning requires deliberate preparation through appropriate financial instruments. Better financial education translates into greater knowledge for this cluster. A policy should focus on resilience instruments, such as unemployment insurance or highly liquid savings, reducing the investment on illiquid assets (e.g., real estate or long-term bonds). This strategy allows an immediate availability of resources when exogenous shocks occur.

Tier 3: The preservation-based intervention for the secure segments: C1 and C2

For the high-performing segments, specifically The Resilient (C2) and The Established (C1), the strategy should focus on wealth preservation and financial literacy upgrades. Since C1 exhibits privileged knowledge of complex and advanced financial concepts, including probabilistic and compound interest calculations, the intervention should focus on leveraging C2 financial abilities to promote an upgrade in their financial profile. Otherwise, given a demographic profile skewed toward older and wealthier individuals, another policy should focus on pension maintenance by facilitating low-fee, fiduciary financial advice to avoid capital erosion, which is fundamental for a healthy retirement period.

6. Conclusion

This research advances the measurement of financial wellbeing by moving from linear, single-score indices to a comprehensive and replicable framework. Through the application of a Random Forest algorithm to a nationally representative sample, financial wellbeing measurement is validated as a complex ecosystem consisting of six distinct profiles, where intertemporal connections between short-term preparedness and long-term security provide significant insights.

Beyond the theoretical novelty of the framework, the primary practical contribution is the identification of the "subjective dominance effect", "imbalance drivers", and the "advanced literacy threshold". These results challenge traditional beliefs by demonstrating that subjective perceptions of personal finances are the core driver behind wellbeing segmentation, rather than material possessions or financial education. This finding suggests that psychological factors act as structural determinants that can either amplify or limit material and cognitive benefits. Furthermore, while general education strengthens long-term security, advanced financial proficiency (mastery of complex instruments) is a prerequisite for accessing top-tier financial profiles, acting as a barrier to entry for the middle class.

Crucially, the distinction between The Short-Sighted (C3) and The Illiquid Planners (C4) resolves a critical ambiguity in the current literature. While both clusters exhibit vulnerability, the underlying drivers are fundamentally different. C3 is constrained by human capital deficits and behavioural myopia, while C4 is destabilised by exogenous shocks. This differentiation provides a data-driven insight for accurate policymaking, shifting from traditional literacy programmes to tailored interventions on structural deficiencies.

The interpretation of these findings requires the acknowledgement of specific methodological constraints. First, the rigorous data pre-processing involves a cleaning procedure that drops incomplete survey responses. While necessary to ensure index consistency, this may introduce survivorship bias, potentially underrepresenting the individuals with fragmented records, who are often the most vulnerable. Second, the cross-sectional nature of the data limits temporal analysis, which would enrich the understanding of movement dynamics within the framework across different stages of individuals' lives. Finally, as the study is based on the U.S. financial infrastructure, generalisations to emerging economies with different social systems should be approached with caution.

Future research should focus on longitudinal panel data to map the dynamics of individuals shifting between clusters over the lifecycle. Additionally, while this study captures the unusual effects of COVID-19 economic disruptions, further research is needed to analyse how these clusters react to other large-scale macro-stressors, such as political or environmental crises. Finally, given the primary significance of subjective perceptions demonstrated in this research, future qualitative analysis could provide deeper insight into the mindset gap between The Established (C1) and The Distressed (C6), promoting educational and psychological mechanisms that may help achieve financial freedom among populations.

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